

MODUL UNIVERSITY VIENNA

SEID 2364: Societal and Ethical Impact of Data Science

SESSION 4

Frameworks

David B. Smith



Opening Provocations

- Write in your private journal
- Stream of consciousness.
- Capture your **immediate intuition**, not a polished or argued answer.
- Do not analyze or justify.
- Words or short phrases are enough.

(Write. Do not discuss yet.)

Provocation

When no option avoids harm, what does it mean to choose the “right” action?

Provocation

- Provide an example where two people arrive at different—but defensible—evaluations of harm in the same situation.

Provocation

What different ways might people
decide between competing harms?



Today's Plan

- workflow (GitHub + Zotero)
- how we make decisions about harm
- Build toward structured decision-making (frameworks)





Workflow Integration

- GitHub → structure and presentation
 - Zotero → research and citation
- 

GitHub Repository folder



[GitHub Assignment Link](#)

In-Class GitHub Tasks

- Convert Assignment 3 to **Markdown** (harm-analysis.md) and upload it to your folder
- Create README.md with your current definitions of **Ethics**, **Intelligence**, and **Harm**
- Commit and push both files to your personal branch

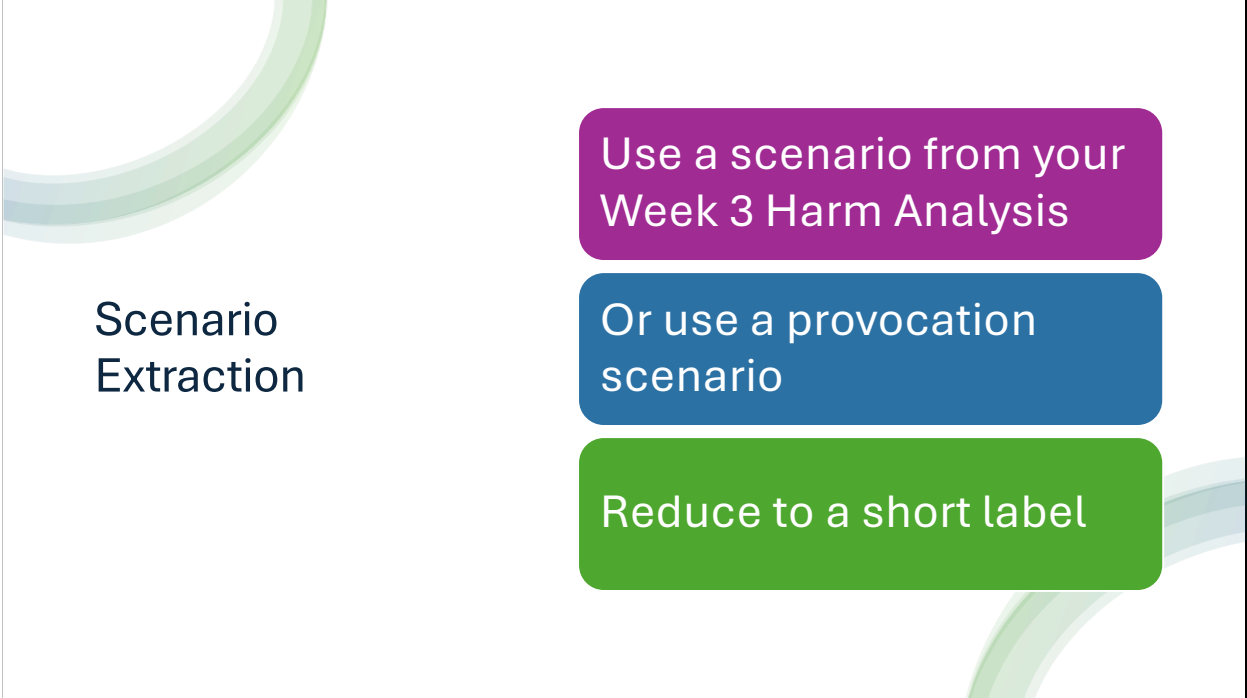
Zotero – Add a Citation.

In-Class Zotero Task

- Add **one scholarly or credible source related to harm** to the **shared Zotero library**
- Create your **own Zotero collection** to organize sources for your emerging research project
- Ensure the citation is complete (metadata + link or attachment)

Return to Your Responses

- We will now examine how we made our decisions.



Scenario Extraction

Use a scenario from your
Week 3 Harm Analysis

Or use a provocation
scenario

Reduce to a short label

WHITE BOARD LAYOUT

Whiteboard Activity (3 Columns)

1. Scenario

System & actors

Decision point

2. Harm

Types of harm

Stakeholders

Conflicts

3. Decision

Core dilemma

Trade-off

X vs Y

Reasoning Examples



Minimize total
harm



Do not
intentionally harm

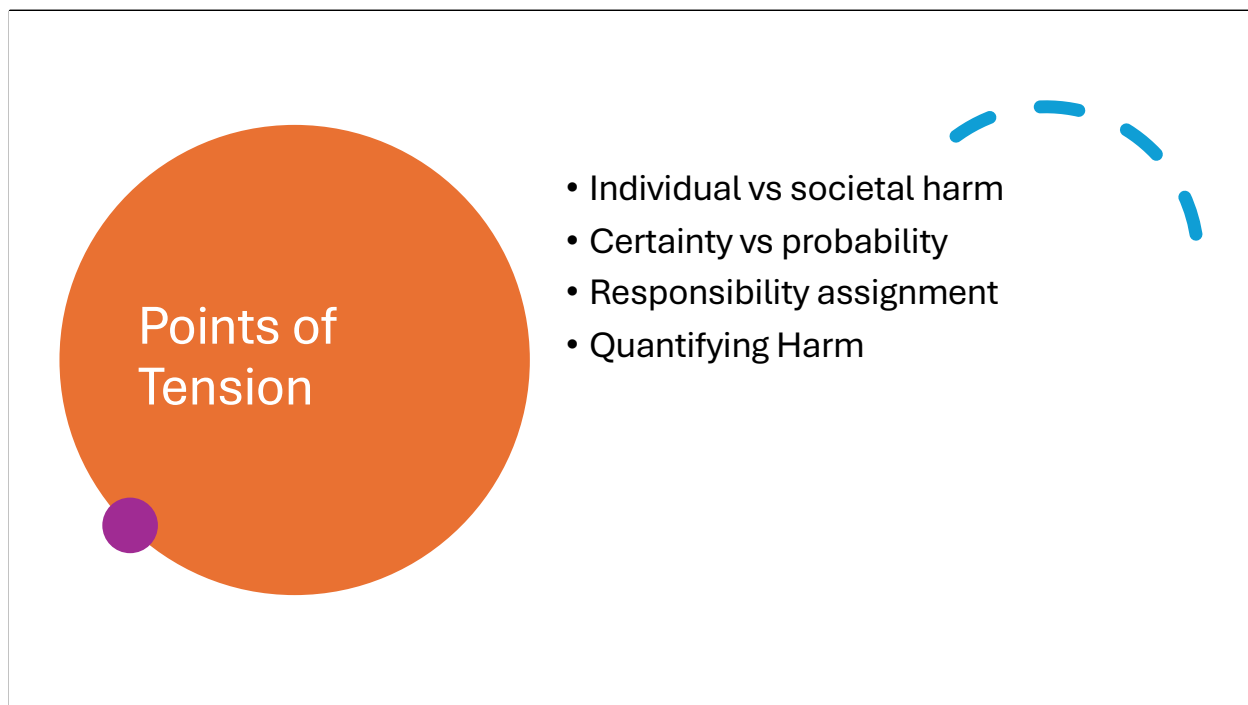


Foreseeability
matters

Three sample claims, each tied to a different framework:

- "Minimize total harm" -> consequentialist reasoning (focus on aggregate outcomes across stakeholders).
- "Do not intentionally harm" -> deontological reasoning (a hard rule/constraint, regardless of outcome).
- "Foreseeability matters" -> virtue/judgment-based reasoning (context, prudence, and what a reasonable agent could anticipate).

Key point: the same situation can produce different verdicts depending on which lens is applied - useful prompt for class discussion before introducing the frameworks formally.



Quantifying harm sounds neutral, but every step hides a value judgment. Walk through the three tensions:

- Individual vs societal harm: do we sum across people, or protect each person as inviolable? Aggregation can mask concentrated harm to a minority.
- Certainty vs probability: a small certain harm vs a large probabilistic one - expected value isn't the only defensible answer (risk aversion, catastrophic tails, irreversibility).
- Responsibility assignment: who bears the harm - the designer, deployer, user, or the affected party? Quantification alone doesn't answer this.

Takeaway: numbers don't resolve these - they reframe them. This motivates the framework-by-framework comparison that follows.

Question

- Are we comparing different kinds of harm, or the same harm applied to different people?

Extending the Scenario: Multi-factor harm

- Death vs injury
- Immediate harm vs long-term societal trust
- Individual harm vs systemic impact

Key Shift

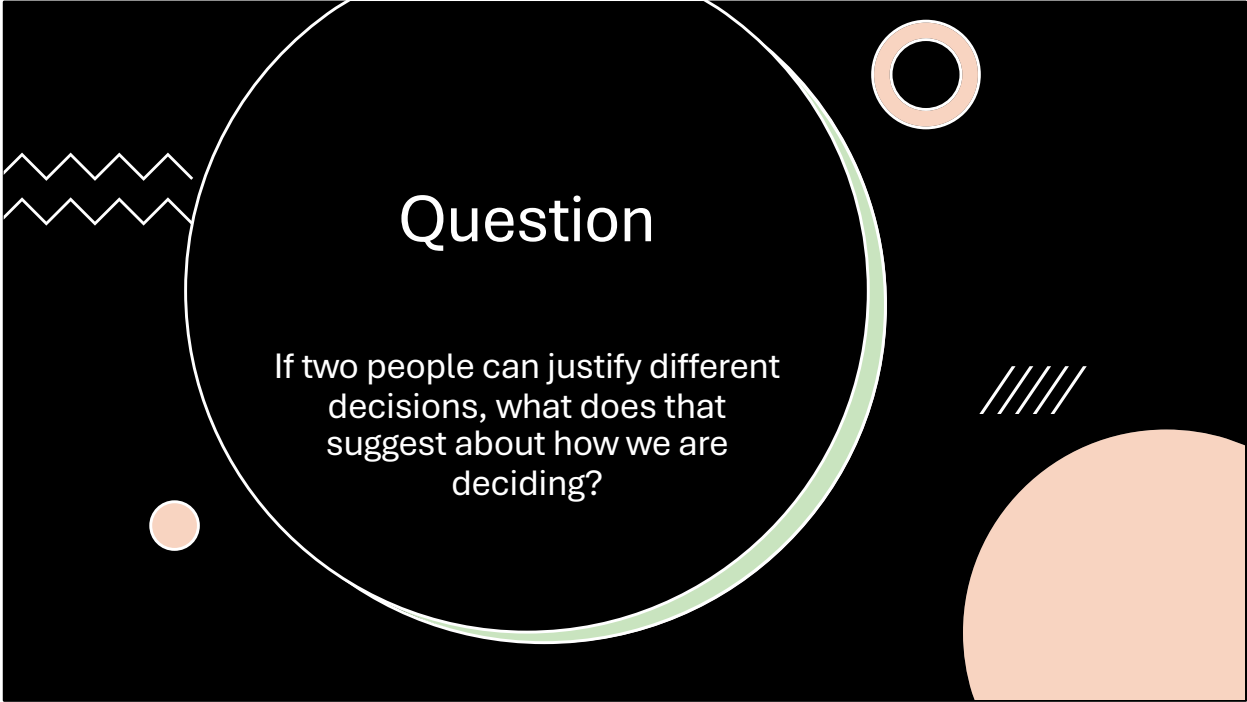


Instability

The background of the slide features a blurred image of a crowd of people, represented by dark silhouettes. The image is split vertically by a white line, with the left side having a blue-to-orange gradient and the right side having an orange-to-red gradient.

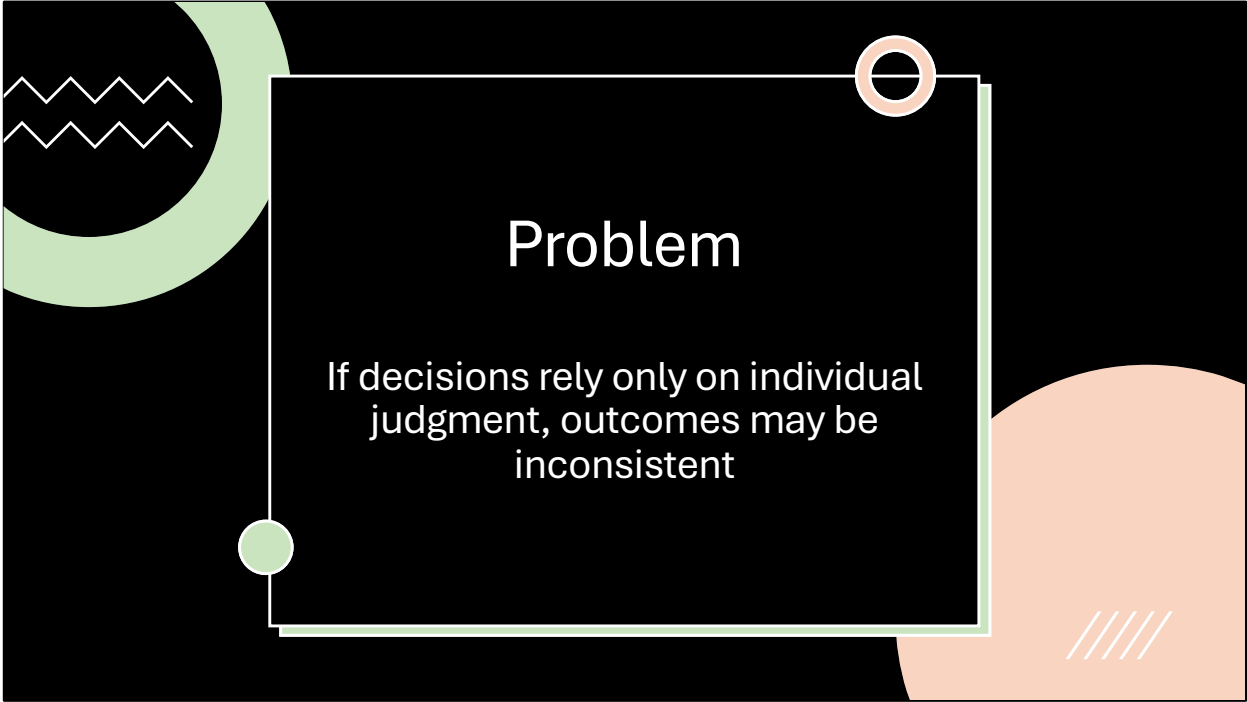
Different people →
different answers

Same situation →
different evaluations



Question

If two people can justify different decisions, what does that suggest about how we are deciding?



Problem

If decisions rely only on individual judgment, outcomes may be inconsistent

Transition



We need a way to make
decisions more consistent

Framework Systems

A framework system is
useful only if it can...?

Possible Criteria

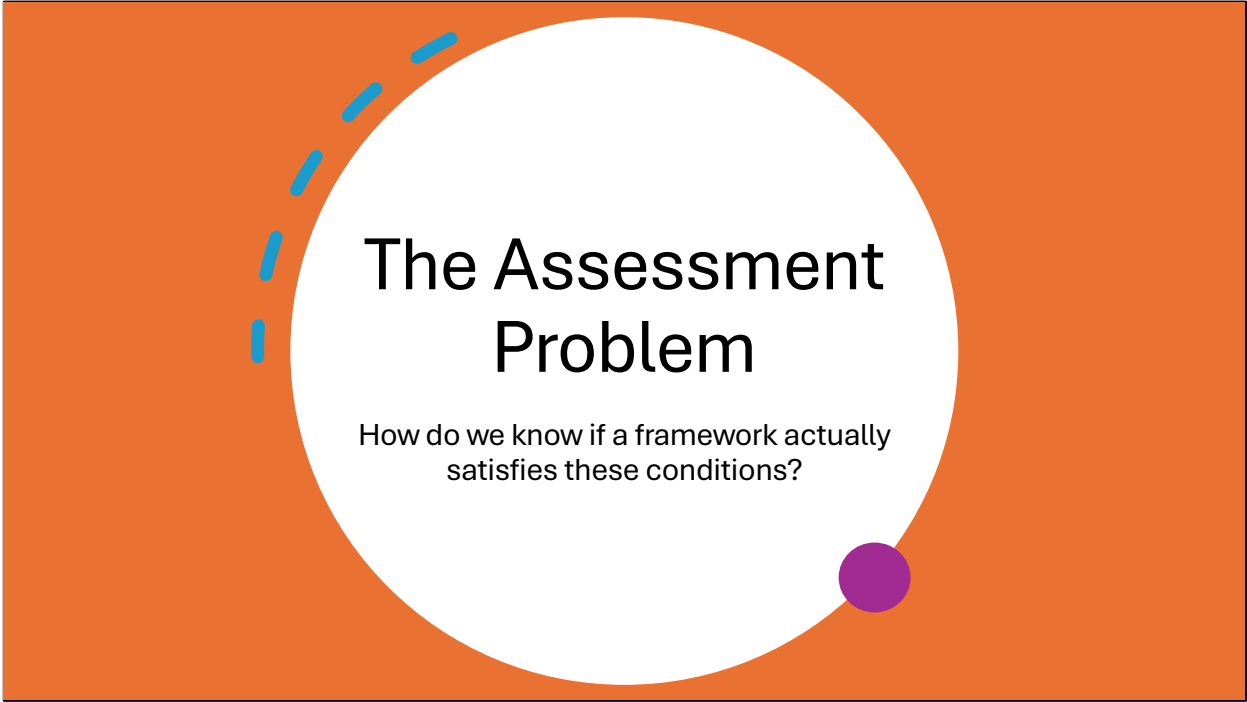
Consistency

Clarity

Fairness

Applicability

Justification



The Assessment Problem

How do we know if a framework actually satisfies these conditions?

Example
Focus:
Fairness

- If a system claims to be fair, how would we determine that?



Harm Types

Physical

Economic

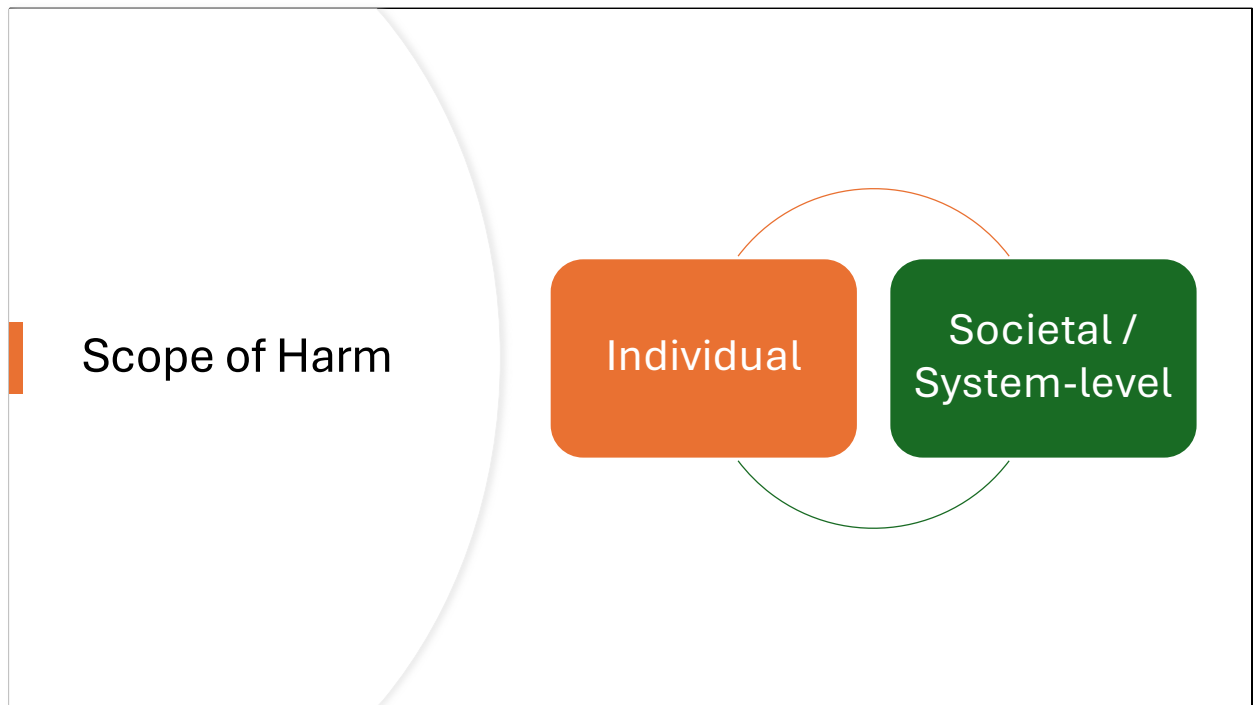
Psychological

Social

Reputational

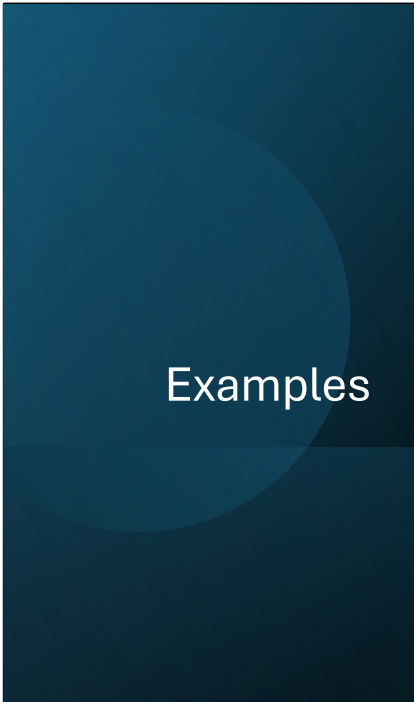
Temporal

*Non-exhaustive List



The Comparison Problem

How do we compare
different types of
harm?



Examples

Physical injury vs financial loss

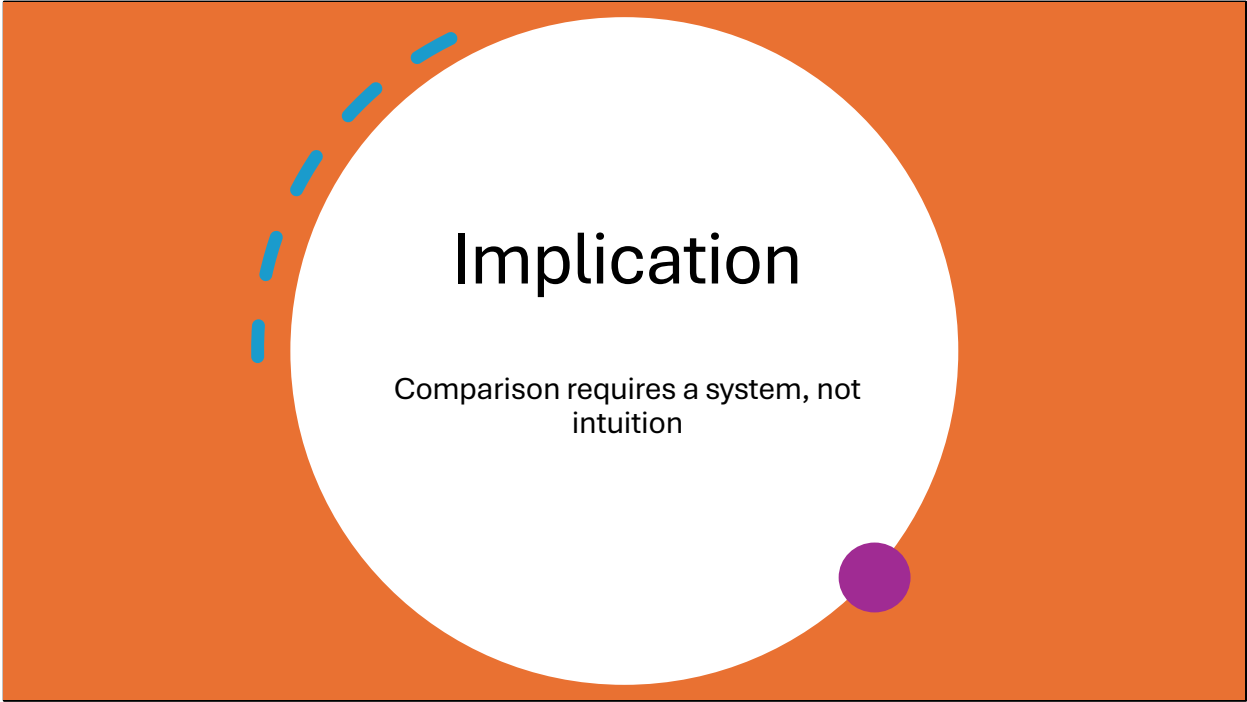
Individual vs societal harm

Immediate vs delayed harm



Key Insight

Harms are not directly comparable

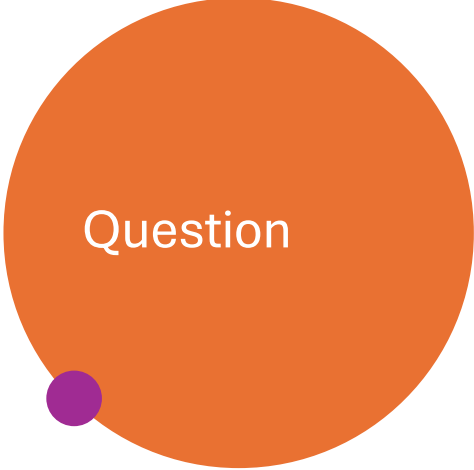


Implication

Comparison requires a system, not
intuition

System Requirements

- Identify parameters
- Measure magnitude
- Compare across types
- Account for scope
- Determine weighting



Question

- Can harm to a group justify harm to an individual?



Philosophical
Frameworks
Exist

Different systems solve these problems in
different ways



Consequentialism

Ethical value depends on the **magnitude and distribution of harms and benefits** produced by an action.



Focus: outcomes Source: Utilitarian tradition (Jeremy Bentham, John Stuart Mill)

Assessment Mechanism:

- Define outcome metrics (types of harm)
- Estimate probabilities and magnitudes - Aggregate (sum/expected value) across stakeholders - Sensitivity to weighting assumptions - Check for omitted harms (indirect, long-term, distributional)



Focus: rules and constraints Source: Kantian ethics (Immanuel Kant)

Assessment Mechanism:

- Identify applicable rules/rights/duties
- - Test for violations (hard constraints)
- - Evaluate universality/consistency of maxims
- - Prioritize constraints over outcomes when in conflict
- - Check for edge cases and rule conflicts (exceptions, thresholds)



Virtue Ethics

Ethical value depends on **judgment and character**, not calculable rules or outcomes.

Focus: judgment and character

Source: Classical Greek philosophy (Aristotle)

Assessment Mechanism: - Identify relevant virtues (e.g., justice, prudence) -

Evaluate context-sensitive judgment (phronesis) - Compare actions to exemplar agents - Balance competing virtues (no fixed weighting) - Check for blind spots (bias, cultural variance, lack of codified rules)



Focus: relationships and responsibility

Source: Feminist ethics (Carol Gilligan, Nel Noddings)

Assessment Mechanism: - Map stakeholders and their relationships - Identify dependencies and vulnerabilities - Evaluate responsibilities owed within those relationships - Center the voices of those most affected - Check for marginalized or unheard parties (power imbalances, relational harms)



- Different frameworks produce different answers





Key Idea

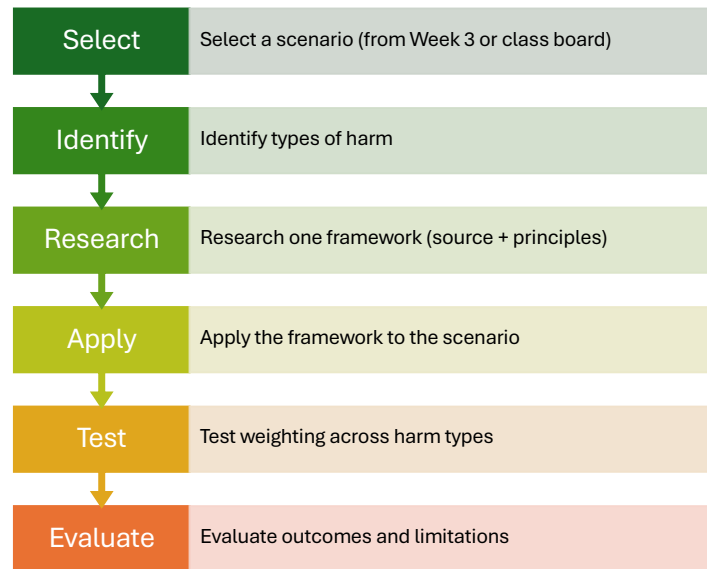
The goal is not agreement, but
structured reasoning

Assignment Preview

You will investigate
and test a
framework

+
o •

Assignment Components



Closing



Different systems → different answers



Structured reasoning matters